

ET 8306 OPTICAL COMMUNICATION SYSTEM

CLASS ASSIGNMENT

08/01/2025

To be submitted on Friday 10/01/2025

The mostly used optical detectors in optical communications are PN – Photodiodes, PIN – Photodiodes, APD– Photodiodes, and MSM – Photodiodes.

- a) Briefly explain the principle of working of each photodetector
- b) Describe the Responsivity, Quantum efficiency, Rise time, and Bandwidth for each photodetector
- c) With calculation examples, give the mathematical formula for calculating Responsivity, Quantum efficiency, Rise time for each photodetector.
- d) Compare the performance of the named photodetectors above.